Shun Yang

Engineer, Guangzhou Marine Geological Survey

Seismic Oceanography | Marine Geophysics | Ocean Mixing

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ResearchGate Profile
GitHub Profile

Research Profile

Shun Yang works on oceanic eddies, thermohaline staircases, ocean fronts, and related phenomena in the Arctic Ocean, offshore Central America, and the Caribbean Sea using seismic reflection methods. His research interests include seismic oceanography, marine geophysics, physical oceanography, mesoscale and submesoscale eddies, thermohaline fine structure, internal waves, ocean mixing, and data visualization.

Current Position

Engineer Current

Guangzhou Marine Geological Survey, Guangzhou, China

Research interests: seismic oceanography, marine geophysics, oceanic eddies, internal waves, thermohaline staircases, and ocean mixing.

Education

Ph.D. in Geophysics Sep 2019 – May 2025

Tongji University

Field of study: Marine Geophysics.

B.Eng. in Exploration Geophysics Sep 2015 – Jun 2019

China University of Geosciences

Field of study: Exploration Geophysics.

Awards and Scholarships

- Outstanding Doctoral Dissertation of Tongji University (同济大学优秀博士学位论文), Dec 2025.
- Outstanding Graduate of Shanghai (上海市优秀毕业生), May 2025.
- Outstanding Student of Tongji University (同济大学优秀学生), Dec 2024.
- National Scholarship for Doctoral Students (博士研究生国家奖学金), Dec 2024.
- Outstanding Doctoral Student Scholarship of Tongji University (同济大学优秀博士生奖学金), Dec 2022.
- Student Excellent Paper Award, Annual Meeting of Chinese Geoscience Union (中国地球科学联合学术年会学生优秀论文奖), Dec 2022.

Publication Summary

Published journal articles	15 articles listed in the local publication file.
First-author journal articles	5 articles, including publications in Communications Earth & Environment, Journal of Geophysical Research: Oceans, Advances in Polar Science, and Chinese Journal of Geophysics.
Additional outputs	Preprints, posters, and conference contributions are retained in a separate section.
Research focus	Seismic oceanography, mesoscale and submesoscale eddies, thermohaline staircases, internal waves, and ocean mixing.

Skills and Expertise

- Physical oceanography; marine geophysics; seismic reflection; seismic oceanography.
- Mesoscale eddies; submesoscale structures; thermohaline staircases; ocean fronts; internal waves.
- Hydrographic analysis, reanalysis and satellite data integration, ocean mixing estimation, and scientific visualization.

Published Journal Articles

1. **Yang Shun**; Zhang Kun; Song Haibin; Ruddick Barry; Liu Mengli; and Meng Linghan. "Disruptions in thermohaline staircases caused by subsurface mesoscale eddies in the eastern Caribbean Sea." *Communications Earth & Environment, 5(1), 1–13*. 2024. https://doi.org/10.1038/s43247-024-01577-3

2. **Yang Shun**; Song Haibin; Coakley Bernard; and Zhang Kun. "Enhanced mixing at the edges of mesoscale eddies observed from high-resolution seismic data in the western Arctic Ocean." *Journal of Geophysical Research: Oceans, 128(10), e2023JC019964*. 2023. https://doi.org/10.1029/2023JC019964
3. **Yang Shun**; Song Haibin; Coakley Bernard; Zhang Kun; and Fan Wenhao. "A mesoscale eddy with submesoscale spiral bands observed from seismic reflection sections in the Northwind Basin, Arctic Ocean." *Journal of Geophysical Research: Oceans, 127(3), e2021JC017984*. 2022. https://doi.org/10.1029/2021JC017984
4. **Yang Shun**; Song Haibin; and Zhang Kun. "Research on submesoscale eddy and front near the South Shetland Islands (Antarctic Peninsula) using seismic oceanography data." *Advances in Polar Science, 33(1), 110–118*. 2022. https://doi.org/10.13679/j.advps.2021.0004
5. 杨顺; 宋海斌; 范文豪; and 吴迪. "中美洲鹦鹉湾气旋涡的亚中尺度结构特征." 地球物理学报*, 64(4), 1328–1340*. 2021. https://doi.org/10.6038/cjg2021O0204
6. Meng Linghan; Zhang Kun; Haibin Song; and **Shun Yang**. "Spatial variation of diapycnal mixing estimated from high-resolution seismic images of subsurface eddies, Bering Sea." *Deep Sea Research Part I: Oceanographic Research Papers, 225, 104601*. 2025. https://doi.org/10.1016/j.dsr.2025.104601
7. Zhang Kun; Song Haibin; Meng Linghan; and **Yang Shun**. "Distribution and characteristics of the subsurface eddies in the Aleutian Basin, Bering Sea." *Journal of Geophysical Research: Oceans, 129(11), e2024JC021402*. 2024. https://doi.org/10.1029/2024JC021402
8. Meng Linghan; Song Haibin; Guan Yongxian; **Yang Shun**; Zhang Kun; and Liu Mengli. "Energy transfer from internal solitary waves to turbulence via high-frequency internal waves: seismic observations in the northern South China Sea." *Nonlinear Processes in Geophysics, 31, 477–495*. 2024. https://doi.org/10.5194/npg-31-477-2024
9. Liu Mengli; Song Haibin; Zhang Kun; **Yang Shun**; and Meng Linghan. "Characteristics of thermohaline staircases in the Southeast Caribbean Sea revealed by seismic reflection data." *Deep-Sea Research Part I: Oceanographic Research Papers, 205, 104246*. 2024. https://doi.org/10.1016/j.dsr.2024.104246
10. 孟令寒; 宋海斌; 梁智超; 范文豪; 刘梦丽; and 杨顺. "加利福尼亚湾温盐阶梯的发现." 地球物理学报*, 66(4), 1634–1648*. 2023. https://doi.org/10.6038/cjg2022Q0231
11. 张鹏; 宋海斌; 范文豪; 杨顺; and 刘梦丽. "加勒比海东南部温盐阶梯的地震海洋学研究." 地球物理学报*, 66(3), 1139–1152*. 2023. https://doi.org/10.6038/cjg2022P0888
12. Zhang Kun; Song Haibin; Coakley Bernard; **Yang Shun**; and Fan Wenhao. "Investigating eddies from coincident seismic and hydrographic measurements in the Chukchi Borderlands, the Western Arctic Ocean." *Journal of Geophysical Research: Oceans, 127(10), e2022JC018453*. 2022. https://doi.org/10.1029/2022jc018453
13. 梁智超; 宋海斌; 范文豪; and 杨顺. "加利福尼亚海湾及其邻近海域亚中尺度现象的地震海洋学研究." 地球物理学报*, 65(8), 3040–3053*. 2022. https://doi.org/10.6038/cjg2022P0509
14. Fan Wenhao; Song Haibin; Gong Yi; **Yang Shun**; and Zhang Kun. "Regional study of mode-2 internal solitary waves at the Pacific coast of Central America using marine seismic survey data." *Nonlinear Processes in Geophysics, 29, 141–160*. 2022. https://doi.org/10.5194/npg-29-141-2022
15. Fan Wenhao; Song Haibin; Gong Yi; Sun Shaoqing; Zhang Kun; Wu Di; Kuang Yunyan; and **Yang Shun**. "The shoaling mode-2 internal solitary waves in the Pacific coast of Central America investigated by marine seismic survey data." *Continental Shelf Research, 212, 104318*. 2020. https://doi.org/10.1016/j.csr.2020.104318

Preprints, Posters, and Conference Contributions

1. Meng, L.; Song, H.; Guan, Y.; **Yang, S.**; Zhang, K.; and Liu, M.. "High-frequency internal waves, high-mode nonlinear waves and K-H billows on the South China Sea's shelf revealed by marine seismic observation." *EGUsphere preprint*. 2024. doi:10.5194/egusphere-2024-92
2. **Yang, S.**; Song, H.; Coakley, B.; and Zhang, K.. "Mesoscale eddies and enhanced mixing at the edges of them observed from high-resolution seismic data in the Western Arctic Ocean." *Poster*. 2023.
3. **Yang, S.**; Song, H.; Coakley, B.; Zhang, K.; and Fan, W.. "A mesoscale eddy with submesoscale spiral bands observed from seismic reflection sections in the Northwind Basin, Arctic Ocean." *Ocean Science Meeting 2022 poster*. 2022. doi:10.13140/RG.2.2.10018.09926
4. Fan, W.; Song, H.; Yi, G.; **Yang, S.**; and Zhang, K.. "Regional study of mode-2 internal solitary waves in the Pacific coast of Central America using marine seismic survey data." *Preprint*. 2021. doi:10.5194/npg-2021-31
5. **Yang, S.**; Song, H.; and Zhang, K.. "A perfect bowl-like submesoscale vortex from seismic reflection transect in the Chukchi Sea, Arctic Ocean." *EGU General Assembly 2021 conference paper*. 2021. doi:10.5194/egusphere-egu21-4760
6. Fan, W.; Song, H.; Zhang, K.; Yi, G.; **Yang, S.**; and Wu, D.. "Using reflection seismic method to estimate the fluid dynamics parameters related to wakes." *EGU General Assembly 2021 conference paper*. 2021. doi:10.5194/egusphere-egu21-9332